

Milestones & Budget Request Breakdown

Grant Project: Riva — Deterministic Performance Diagnostics Companion for Unreal Engine

Requested Amount: \$50,000 USD

Duration: 6 Months (4 Technical Milestones)

Technical Milestones Roadmap

Month 1-2		Month 3		Month 4-5		Month 6	
■ Milestone 1	■	■ Milestone 2	■	■ Milestone 3	■	■ Milestone 4	■
■ Extended .utrace	■	■ Call-stack	■	■ Standalone Desktop	■	■ Fab Marketplace	■
■ Subsystem Extraction	■	■ Symbolication	■	■ GUI Application	■	■ & GitHub Actions	■
■ (\$15,000)	■	■ (\$15,000)	■	■ (\$10,000)	■	■ (\$10,000)	■

Milestone 1: Extended Subsystem Extraction & Advanced UE5 Signatures

- **Duration:** Months 1 – 2
- **Budget:** \$15,000 USD
- **Deliverables:**
 - Expand FRivaTraceService to extract deep TraceServices markers for **World Partitioning** grid loading, **Chaos Physics** solver steps, and **Niagara VFX** compilation lanes.
 - Implement 5 new built-in core signatures (STUT_WORLD_PARTITION, STUT_CHAOS_PHYSICS, STUT_NIAGARA_COMPILE, STUT_ANIMATION_EVAL, STUT_TEXTURE_STREAMING).
 - Add CTest suite verification for all new subsystem signatures.

Milestone 2: Call-Stack Symbolication & Native Stack Unwinding

- **Duration:** Month 3
- **Budget:** \$15,000 USD
- **Deliverables:**
 - Integrate native symbol resolution for call-stacks embedded in .utrace recordings (PDB on Windows, dSYM on macOS, ELF/DWARF on Linux).
 - Surface function-level flamegraph evidence directly inside the SRivaPanel Slate details pane and Markdown reports.
 - Ensure zero external cloud dependencies during offline symbol resolving.

Milestone 3: Cross-Platform Standalone Desktop Companion App

- **Duration:** Months 4 – 5
- **Budget:** \$10,000 USD
- **Deliverables:**
 - Develop a standalone desktop GUI companion application (`riva-gui`) using C++20 and Dear ImGui / Slate for developers who want to inspect traces outside of Unreal Editor.
 - Support drag-and-drop trace file loading, live trace comparison side-by-side split view, and direct export.
 - Distribute pre-built native binaries for Windows, macOS, and Linux.

Milestone 4: Fab Marketplace Integration & CI/CD Tooling Package

- **Duration:** Month 6
- **Budget:** \$10,000 USD
- **Deliverables:**
- Package RivaEditor for official distribution on Epic Games' **Fab Marketplace**.

- Publish an official GitHub Action (riva-action) and Unreal Automation Tool (UAT) plugin script for zero-config CI/CD pipeline integration.
 - Create video walkthroughs and comprehensive developer documentation.
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Summary of Funds Allocation

Expense Category	Allocation	Percentage	Description
Core C++ Development	\$30,000	60%	Full-time engineering for TraceServices extraction, symbolication, and signature algorithms.
GUI & Editor UX Design	\$10,000	20%	Standalone desktop application development & Slate UI refinements.
CI/CD Tooling & Distribution	\$10,000	20%	GitHub Action creation, Fab Marketplace packaging, video tutorials, and documentation.
Total	\$50,000	100%	